

Anvaya

A toolkit for phylogenetic analysis of chromosome fusion-and-mixing characters

User tutorial and worked examples

What this document covers

A complete, practical guide to running Anvaya: what it does, when each mode applies, how to choose every threshold from your own data rather than by guesswork, and eight fully worked case studies with runnable scripts. Examples are drawn from across animals and plants — teleost fish, frogs, nematodes, molluscs and land plants as well as chelicerates — because the method is not clade-specific.

Read Section 2 first

Anvaya's single most consequential requirement is that genomes carrying a lineage-specific whole-genome duplication must not be used to define linkage groups. Section 2 explains why, and gives a reference table of which common model lineages are and are not duplicated. Getting this wrong degrades the analysis silently rather than producing an error.

Contents

§	Section	Page
1	What Anvaya does	3
2	Duplicated vs. unduplicated genomes — read this first	4
3	Installing and checking your inputs	6
4	The character model	8
5	Choosing a mode: reference vs. consensus	9
6	Tuning the consensus thresholds	10
7	Worked examples	12
7.1	Example A — you already have .rbh files (molluscs vs. BCnS ALGs)	12
7.2	Example B — raw genomes, single reference (nematodes)	12
7.3	Example C — multi-genome consensus (insects)	12
7.4	Example D — a duplicated lineage: teleost fish vs. spotted gar	14
7.5	Example E — allopolyploidy: Xenopus laevis vs. X. tropicalis	15
7.6	Example F — testing a contested node with a backbone constraint	15
7.7	Example G — publication figures	16
7.8	Example H — full sensitivity analysis	16
8	Reading the output	18
9	Troubleshooting	20
10	Complete flag reference	21

1. What Anvaya does

Chromosome-scale blocks of orthologous genes — **ancestral linkage groups** (ALGs) — persist across enormous evolutionary distances. Because they are rearranged by rare chromosomal events rather than by nucleotide substitution, they supply phylogenetic characters that are largely independent of sequence alignment, and comparatively immune to long-branch and compositional artefacts.

Anvaya codes two such events as discrete characters and runs a parsimony analysis on them:

- **Fusion** — two ancestral linkage groups now sharing one chromosome.
- **Mixing** — the genes of two fused groups having become interleaved rather than remaining as two contiguous blocks. Mixing is widely argued to be effectively irreversible, which makes it a polarity-known character.

The toolkit is four Python scripts:

Script	Role
`genomes_to_rbh.py`	Raw genomes → `.rbh` ortholog tables. Runs DIAMOND reciprocal best hits, defines linkage groups, filters short scaffolds, rescues duplication-derived orthologs.
`alg_parsimony.py`	`.rbh` → character matrix, parsimony trees, ancestral states, synapomorphies, support statistics, figures, NEXUS/TNT export.
`sweep_consensus_params.py`	Sweeps the three consensus thresholds and writes one summary TSV to guide parameter choice.
`make_toy_dataset.py`	Generates a synthetic dataset with known ground truth for testing the pipeline.

Two optimality criteria are always run side by side on the same characters, so that any difference between the resulting trees is attributable purely to the reversibility assumption:

- **Reversible** — ordinary unordered (Fitch/Sankoff) parsimony; states may change in either direction.
- **Irreversible** — states may only increase (0→1→2), a Camin-Sokal model encoding the claim that fusion and mixing do not spontaneously revert. Requires a rooted tree.

2. Duplicated vs. unduplicated genomes — read this first

Anvaya establishes what counts as a linkage group either from one **reference** genome or from a set of **core** genomes. Whichever you choose, those genomes must not carry a whole-genome duplication that the rest of your sample does not share.

2.1 Why it matters

Reciprocal best hit is a strict 1:1 mapping. A duplicated genome breaks that assumption in two different, both damaging, ways depending on the mode:

- **As a reference:** each true ancestral chromosome is present twice, so it is registered as *two* separate linkage groups. Every non-duplicated taxon then appears to be missing half of each one — which is a ploidy mismatch, not biology, but is indistinguishable from real signal downstream.
- **As a core genome:** orthogroups containing more than one gene from the same core genome are discarded as ambiguous. A duplicated core genome therefore causes its ohnologous orthogroups to vanish silently, gutting the reconstruction with no error message.

The rule

Duplicated lineages belong on the *comparison* side of the analysis, never in `--reference-species` or `--consensus-species`. Flag them with `--detect-wgd --wgd-species …` so their dropped ohnologs are actively recovered instead.

2.2 “Duplicated” is relative to your sample, not absolute

This is the point most often missed. What matters is not whether a genome has *ever* undergone WGD, but whether it has undergone one that the other genomes in your analysis *did not also inherit*.

All vertebrates share the two ancient rounds (2R) at the base of the group. If every taxon in your study is a vertebrate, 2R is shared ancestry and causes no asymmetry — any vertebrate may serve as reference. But add an invertebrate outgroup such as amphioxus and 2R becomes a lineage-specific duplication *relative to that outgroup*, and the same considerations apply. Always ask: duplicated *relative to what*?

2.3 Reference table

Lineages commonly used in macrosynteny work. “Unduplicated” here means no lineage-specific WGD beyond what its usual comparators share.

Lineage	Status	Notes for Anvaya
Nematodes (*C. elegans*, *Pristionchus*)	Unduplicated	Excellent core/reference taxa. Nigon elements are a well-curated ALG reference.
Insects (*Drosophila*, *Tribolium*, *Apis*)	Unduplicated	Good core taxa; Muller elements are a classic linkage-group system.
Ticks and mites (Acari)	Unduplicated	Safe as core genomes.
Sea spiders (Pycnogonida)	Unduplicated	Confirmed by chromosome-level assembly; sister to all other chelicerates, so an excellent deep outgroup and core taxon.
Amphioxus (*Branchiostoma*)	Unduplicated	Standard pre-2R chordate outgroup and reference.

Lineage	Status	Notes for Anvaya
Molluscs, annelids (most)	Unduplicated	Generally safe; check the specific species, as scattered WGDs exist.
Spotted gar (*Lepisosteus*)	Unduplicated relative to teleosts	The standard non-3R outgroup for teleost work — ideal reference.
Teleost fish (zebrafish, medaka)	**Duplicated** (3R)	Comparison side only. Use `--detect-wgd`.
Salmonids (salmon, trout)	**Duplicated** (3R + 4R)	Two rounds; expect partial rescue only (see §2.4).
Xenopus laevis	**Duplicated** (allotetraploid)	Comparison side. *X. tropicalis* is the diploid congener to use as reference.
Arachnopulmonates (spiders, scorpions, whip scorpions)	**Duplicated** (1 shared WGD)	Comparison side. Non-arachnopulmonate arachnids are unduplicated.
Horseshoe crabs (Xiphosura)	**Duplicated** (multiple)	Comparison side; higher-order redundancy, partial rescue only.
Most crop plants (wheat, soy, *Brassica*)	**Duplicated**	Polyploidy is pervasive in plants; verify per species before choosing core taxa.
Amborella trichopoda	Unduplicated relative to core angiosperms	The standard non-duplicated angiosperm reference.

2.4 What the duplication rescue can and cannot do

`--detect-wgd` self-compares a genome, identifies ohnologous chromosome pairs from intragenomic paralogy, and adds back the ohnolog that plain reciprocal best hit dropped. It was designed and validated for a **single, clean, pairwise** duplication.

- **Works well:** one WGD round, one ohnologous partner per chromosome — teleosts vs. gar, *X. laevis* vs. *X. tropicalis*, arachnopulmonates vs. other arachnids.
- **Partial:** multiple rounds (salmonids, horseshoe crabs) or heavily rearranged post-WGD genomes, where one modern chromosome carries material from several ancestral ohnologs. Rescue still improves substantially on the plain-RBH false negative, but treat the result as a lower bound rather than complete recovery, and inspect `duplication_blocks.tsv` directly.
- **Not handled:** duplication private to the reference or core genomes themselves — which is exactly why §2.1's rule exists.

3. Installing and checking your inputs

3.1 Requirements

Component	Needed for
`Python 3 + pandas`	Everything (only hard dependency)
`diamond on PATH`	genomes_to_rbh.py orthology search
`matplotlib`	All --plot-* figures (optional; skipped with a warning)
`scipy`	Fisher's exact tests (optional; falls back to count-only thresholds)
`biopython`	CDS→protein translation when no .pep is supplied (optional)
`paup / tnt on PATH`	Only for --run-paup / --run-tnt (optional)

3.2 Input layout

One directory, with per-species files sharing a common stem. Annotations may be `.gff3`, `.gff` or `.gtf` — species may mix formats freely.

```
Data/
  Danio.cds     Danio.gff3    Danio.pep      # .pep optional; translated if absent
  Lepisosteus.cds  Lepisosteus.gtf
  Amia.cds      Amia.gff
```

3.3 Check assembly contiguity before you start

A fragmented core genome inflates the linkage-group count no matter how thresholds are set. Check every candidate before committing. Note the extension handling — assuming a single extension silently produces empty output for species using another.

```
for sp in Danio Lepisosteus Amia; do
  echo "=== $sp ==="
  f=$(ls Data/${sp}.gff3 Data/${sp}.gff Data/${sp}.gtf 2>/dev/null | head -1)
  awk -F'\t' '!/^#/{if($5>max[$1]) max[$1]=$5} END{for (s in max) print max[s], s}' "$f" \
    | sort -rn | head -5
done
```

You are looking for a handful of chromosome-scale scaffolds, not a long tail of small fragments. A genome whose largest scaffolds are orders of magnitude smaller than its peers' is a poor core candidate (though still perfectly usable as a comparison taxon).

3.4 Verify the pipeline works

Before touching real data, run the synthetic dataset. It has known ground truth documented in the `MANIFEST.md` it writes.

```
python3 make_toy_dataset.py --outdir toy_data
python3 make_toy_dataset.py --verify-only      # checks sequence separation only

python3 genomes_to_rbh.py --input-dir toy_data --reference-species T1 \
```

```
    --species T1 T2 T3 T4 T5 T6 \
    --rbh-outdir toy_rbh \
    --parsimony-script "$(pwd)/alg_parsimony.py" \
    --parsimony-outdir toy_parsimony \
    --forward --min-genes 3 --alpha 0.05 --outgroup T2 \
      --bootstrap 100 --permutation-test 100 --plot-trees
```

Note the outgroup

T1 is the *reference* here, so it is deliberately not a taxon in the tree and cannot also be the outgroup — T2 is used instead. In consensus mode, core species *are* taxa and can serve as the outgroup. Anvaya reports this explicitly if you get it wrong.

4. The character model

For every pair of linkage groups that co-occur on one chromosome in at least one taxon, each taxon receives one of four states:

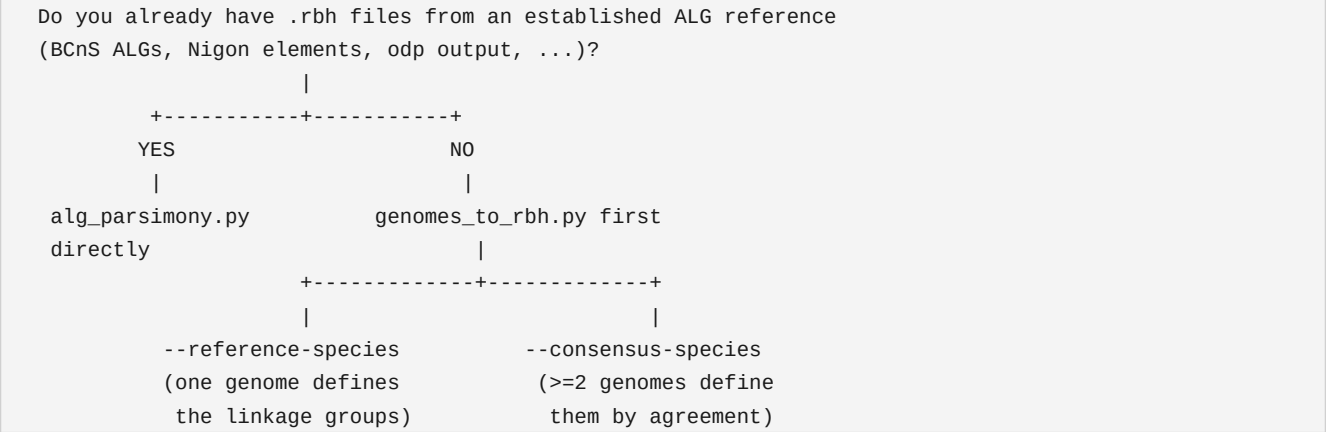
State	Meaning	How it is called
0	Unfused	The two groups sit on different chromosomes.
1	Fused, clean	Co-located, but still two contiguous blocks — the fraction of adjacent gene pairs crossing between groups is at or below `--mixing-threshold`.
2	Fused, mixed	Co-located and interleaved — that fraction exceeds the threshold.
?	Missing	Too few orthologs recovered for either group to assess. Coded missing rather than assumed unfused.

An ALG counts as present on a chromosome when it is supported by at least `--min-genes` orthologs and, where a Fisher's exact enrichment p-value is available, clears `--alpha`. A chromosome carrying three or more linkage groups is decomposed into all pairwise characters, following convention — note that such characters are not strictly independent.

Interpreting mixing

Clean fusion says two chromosomes joined. Mixed fusion says they joined *and* enough time and rearrangement have passed to interleave their genes. Under the irreversible criterion 0→1→2 is a one-way ratchet, which is what gives these characters their polarity.

5. Choosing a mode: reference vs. consensus



5.1 Single-reference mode

One genome's chromosomes *are* the linkage groups. Fast — one bidirectional search per other taxon.

Use when	Avoid when
A well-assembled, unduplicated genome exists whose karyotype is close to ancestral (spotted gar for teleosts, *X. tropicalis* for *X. laevis*, amphioxus for chordates).	No single genome is clearly a good proxy, or the obvious candidate is fragmented or heavily rearranged — its idiosyncrasies become your ALG definitions.

The reference is not a taxon

It has nothing to be compared against, so it gets no `.rbh` file and never appears in the tree — exactly as an external ALG set would not. It therefore cannot be the `--outgroup` or appear in a `--constraint`.

5.2 Multi-genome consensus mode

Linkage groups are reconstructed *de novo*: orthogroups are built across the core genomes, then two orthogroups are placed in the same linkage group when they remain co-located in enough of them. No pre-existing ALG reference is needed, and every genome — core included — becomes a taxon in the tree.

- **Choose core genomes that are phylogenetically independent.** Several close relatives will agree with each other through recent common ancestry, making the agreement check circular.
- **Straddle the ingroup boundary.** If every core genome is an ingroup member, any fusion shared by the whole ingroup is co-located in all of them, reconstructed as a single linkage group, and thereby erased as a character — losing precisely the deep synapomorphies you want. Include at least one outgroup genome to preserve the ancestral state.
- **Never include a duplicated genome** (§2).
- **More core genomes is not automatically better.** Each additional deeply-diverged lineage is another chance for a private rearrangement to block a merge under unanimous agreement.

6. Tuning the consensus thresholds

Three thresholds control the reconstruction. They do genuinely different jobs and should be tuned in this order. None has a universally correct value — all should be read off your own data.

6.1 The fast way: sweep all three at once

```
python3 sweep_consensus_params.py \
    --input-dir ./Data \
    --consensus-species Amborella Vitis Oryza Arabidopsis \
    --outdir ./sweep \
    --min-species 3 4 \
    --min-frac 1.0 0.75 0.5 \
    --min-alg-orthogroups 1 5 10 15 20 30 \
    --threads 30
```

This writes one TSV with a row per combination. It is much faster than it looks: `--min-alg-orthogroups` is a pure post-reconstruction filter, so the whole axis is derived arithmetically from a single reconstruction; and all runs share one work directory, so DIAMOND alignments are computed once for the entire sweep.

Column	What to look for
`n_linkage_groups`	The number you are trying to make biologically plausible. Compare against known haploid chromosome counts for your clade.
`pct_orthogroups_retained`	How much of your data ends up inside a kept group. A combination with an attractive group count but low retention is a bad trade.
`largest_group`	If one group holds a large fraction of all orthogroups, `--consensus-min-frac` is too permissive and unrelated groups are merging.
`median_group`	A healthy reconstruction has a substantial median, not a median of 1–2 with one huge outlier.

6.2 --consensus-min-species

How many core genomes must contain an orthogroup for it to be used. Default: all of them. Watch the log line `N/M orthogroups present in >= K core species`. Across deep divergences, gene loss and annotation gaps compound, and requiring every genome can discard a large fraction of usable data. Allowing one lineage to be missing often recovers a great deal at little cost.

6.3 --consensus-min-frac

What fraction of the core genomes sharing two orthogroups must agree they are co-located before the two are merged. Default 1.0 (unanimous). In clades with real rearrangement, unanimity means one lineage's private translocation blocks a genuine ancestral group, shattering the reconstruction into fragments.

Only k/n fractions are achievable — mind the boundary

Agreement is (genomes agreeing) / (genomes sharing both), so only discrete fractions occur. A threshold sitting just *above* one of them silently behaves like the next one up: with 3 core genomes, `0.67` **rejects** 2-of-3 agreement, because 2/3 = 0.6667 < 0.67 — making it indistinguishable from requiring unanimity. Use `0.66` for 2-of-3. Anvaya prints what your threshold actually means in k-of-n terms on every run, and warns when it sits in one of these traps.

Sweep it and watch for a **smooth, monotonic** decrease in the raw group count. An abrupt collapse toward 1 means unrelated groups are being merged; back off to a stricter value.

> **Beware non-independent core taxa at low thresholds**
>
> With four core genomes, `0.5` means 2-of-4 suffices — so two closely related core genomes can carry a merge over the objections of both other lineages. If your core set contains a pair of close relatives, keep the threshold high enough that they cannot outvote the rest on their own.

6.4 --consensus-min-alg-orthogroups

The other two govern *how* groups form; this one simply discards groups smaller than a given size. Every run writes `consensus_linkage_group_sizes.tsv`, ranking all candidates by member count, so the cut-off can be read from the data:

```
sort -k2,2 -rn rbh_diag/consensus_linkage_group_sizes.tsv | head -40
```

- **If there is a clear cliff** — a cluster of large groups, then a sharp drop to a tail of small ones — set the threshold just above the drop.
- **If the distribution decays smoothly**, which is common for phylogenetically broad samples, no threshold is objectively correct. Do not keep hunting for a knee that is not there.

In the smooth case: choose a value giving a biologically plausible group count, state it in your write-up as an explicit analytical decision rather than a data-derived one, and run the downstream analysis at several thresholds to check whether your conclusions are stable. If they are, the exact cut-off does not matter. If they are not, that instability is itself a finding and belongs in the paper. Because the thresholds interact, vary `min-frac` and `min-alg-orthogroups` together, not one at a time.

7. Worked examples

Each example is self-contained and runnable after substituting your own file names.

7.1 Example A — you already have .rbh files

Scenario: molluscan genomes already compared against the BCnS ALG set by an odp-style pipeline. No orthology work is needed; go straight to the phylogenetic analysis.

```
python3 alg_parsimony.py \
    --indir ./rbh_files \
    --outdir ./results \
    --min-genes 3 --alpha 0.05 \
    --outgroup Lingula \
    --plot-dotplots --plot-trees --plot-matrix \
    --bootstrap 100 --permutation-test 100 \
    --export-nexus --export-tnt
```

The shared reference column prefix (`BCnSSimakov2022_*` or similar) is detected automatically; override with `--reference-prefix` if needed. Always run the permutation test before interpreting a tree — it asks the prior question of whether the matrix carries signal at all.

7.2 Example B — raw genomes, single reference

Scenario: several nematode genomes. Nematodes are unduplicated, and their chromosomes correspond to well-characterised Nigon elements, so one well-assembled genome is a sound reference.

```
#!/usr/bin/env bash
set -euo pipefail

python3 genomes_to_rbh.py \
    --input-dir ./Data \
    --reference-species Celegans \
    --min-scaffold-length 100000 \
    --threads 30 \
    --rbh-outdir ./rbh_out \
    --parsimony-script "$(pwd)/alg_parsimony.py" \
    --parsimony-outdir ./parsimony_out \
    --forward \
      --min-genes 3 --alpha 0.05 \
      --outgroup Pristionchus \
      --plot-dotplots --plot-trees --plot-ideograms \
      --bootstrap 100 --permutation-test 100
```

Everything after `--forward` is passed through to `alg_parsimony.py` unchanged. *C. elegans* is the reference and so is not a taxon — the outgroup must be one of the compared species.

7.3 Example C — multi-genome consensus

Scenario: insect genomes with no curated ALG set at the depth of interest. All are unduplicated, so all are eligible as core genomes. The core set spans independent orders and includes an outgroup so that ingroup-wide fusions remain visible as characters.

```
#!/usr/bin/env bash
```

```
set -euo pipefail

CORE="Drosophila Tribolium Apis Bombyx"

# Step 1 - diagnostic run, everything permissive
python3 genomes_to_rbh.py --input-dir ./Data --consensus-species $CORE \
    --rbh-outdir ./rbh_diag --rbh-only --threads 30 \
    2>&1 | grep -E "orthogroups|reconstructed|min-frac"

sort -k2,2 -rn ./rbh_diag/consensus_linkage_group_sizes.tsv | head -40

# Step 2 - sweep the interacting thresholds
python3 sweep_consensus_params.py --input-dir ./Data --consensus-species $CORE \
    --outdir ./sweep --min-species 3 4 --min-frac 1.0 0.75 0.5 \
    --min-alg-orthogroups 1 5 10 20 30 --threads 30

# Step 3 - full analysis at the chosen setting
python3 genomes_to_rbh.py \
    --input-dir ./Data --consensus-species $CORE \
    --consensus-min-species 3 \
    --consensus-min-frac 0.75 \
    --consensus-min-alg-orthogroups 15 \
    --min-scaffold-length 100000 --threads 30 \
    --rbh-outdir ./rbh_out \
    --parsimony-script "$(pwd)/alg_parsimony.py" \
    --parsimony-outdir ./parsimony_out \
    --forward --min-genes 3 --alpha 0.05 --outgroup Bombyx \
      --plot-dotplots --plot-trees --plot-ribbon \
      --ribbon-order-from-tree reversible \
      --bootstrap 100 --permutation-test 100
```

7.4 Example D — a duplicated lineage: teleosts vs. spotted gar

*Scenario: the textbook case. Teleost fish underwent the 3R whole-genome duplication; spotted gar (*Lepisosteus*) diverged before it and is the standard non-duplicated outgroup. Gar defines the linkage groups; the teleosts are compared against it with ohnolog rescue enabled.*

Step 1 — calibrate the duplication threshold

`--wgd-min-paralog-pairs` counts intragenomic paralog pairs per chromosome pair. Run it effectively off first so the full distribution is visible:

```
python3 genomes_to_rbh.py --input-dir ./Data \
    --reference-species Lepisosteus \
    --detect-wgd --wgd-species Danio Oryzias Takifugu \
    --wgd-min-paralog-pairs 1 \
    --rbh-outdir ./rbh_wgd_diag --rbh-only --threads 30

# counts are in field 4 (field 1 is the taxon) - sort on the right column
sort -k4,4 -rn rbh_wgd_diag/duplication_blocks.tsv | head -40

# how many candidate pairs at each support level?
awk 'NR>1{print $4}' rbh_wgd_diag/duplication_blocks.tsv | sort -rn | uniq -c
```

Real ohnologous chromosome pairs are typically supported by dozens to hundreds of paralog pairs; incidental paralogy sits far lower. Set the threshold just above that noise floor. Two sanity checks: the number of confirmed pairs should be roughly one partner per ancestral chromosome, and for a single clean WGD each chromosome should have one dominant partner.

Step 2 — the full run

```
python3 genomes_to_rbh.py \
    --input-dir ./Data \
    --reference-species Lepisosteus \
    --detect-wgd --wgd-species Danio Oryzias Takifugu \
    --wgd-min-paralog-pairs 25 \
    --min-scaffold-length 100000 --threads 30 \
    --rbh-outdir ./rbh_out \
    --parsimony-script "$(pwd)/alg_parsimony.py" \
    --parsimony-outdir ./parsimony_out \
    --forward --min-genes 3 --alpha 0.05 --outgroup Amia \
      --plot-dotplots --plot-ribbon --highlight-wgd \
      --bootstrap 100 --permutation-test 100
```

Then inspect `parsimony_out/dotplots/Danio.wgd.pdf`. With rescue working you should see ringed, double-diameter points forming paired diagonals across ohnologous chromosomes. Sparse or scattered points mean the threshold needs revisiting.

Why this matters

Without rescue, a duplicated genome's orthologs are split across both ohnologous chromosomes, so each may fall below `--min-genes` and the linkage group is called absent on *both* — the duplication becomes entirely invisible rather than obvious. Rescue reverses that false negative.

7.5 Example E — allopolyploidy: *Xenopus laevis*

Scenario: X. laevis *is allotetraploid; its diploid congener* X. tropicalis *is the natural reference. The two subgenomes (L and S) are the ohnologous copies.*

```
python3 genomes_to_rbh.py \
    --input-dir ./Data \
    --reference-species Xtropicalis \
    --detect-wgd --wgd-species Xlaevis \
    --wgd-min-paralog-pairs 30 \
    --rbh-outdir ./rbh_out --rbh-only --threads 30

cat rbh_out/duplication_blocks.tsv
```

Because allotetraploidy is recent and subgenomes are well preserved, this is close to the best-case scenario for the rescue mechanism — expect clean, high-support chromosome pairs corresponding to the L/S subgenome correspondence.

Reading the duplication report

`duplication_blocks.tsv` lists confirmed ohnologous chromosome pairs with their supporting paralog-pair counts and p-values. For a clean, recent WGD expect roughly disjoint pairs. A connected network instead — chromosomes with several partners — indicates post-WGD rearrangement or higher-order duplication, in which case rescue is partial (§2.4).

7.6 Example F — testing a contested node

Scenario: one node is genuinely contested, while surrounding relationships are well established. Constrain the uncontroversial parts as a positive control and let the linkage-group characters resolve the rest.

Write a Newick constraint. It may name any subset of taxa, and polytomies leave relationships within a group unresolved:

```
cat > constraint.nwk <<'EOF'
((TaxonA,TaxonB,TaxonC),(TaxonD,TaxonE),TaxonF,TaxonG);
EOF

python3 alg_parsimony.py \
    --indir ./rbh_out --outdir ./constrained \
    --min-genes 3 --alpha 0.05 --outgroup TaxonG \
    --constraint constraint.nwk \
    --bootstrap 100 --permutation-test 100 \
    --export-nexus --export-tnt
```

Each internal node asserts that its full descendant set is monophyletic. Taxa the constraint omits may attach anywhere. Both criteria honour the constraint identically — note that the output names `tree_reversible` / `tree_irreversible` refer to the *character model*, not to the presence of a backbone constraint.

Use a positive control

Constrain a clade you are confident about and check that the characters recover it unconstrained too. If they cannot recover a well-established group on their own, that tells you how much weight the contested node can bear.

7.7 Example G — publication figures

```
python3 alg_parsimony.py \
    --indir ./rbh_out --outdir ./figures \
    --min-genes 3 --alpha 0.05 --outgroup Lepisosteus \
    --plot-dotplots \
    --plot-ideograms \
    --plot-matrix \
    --plot-trees \
    --plot-ribbon --ribbon-order-from-tree reversible \
    --highlight-wgd \
    --plot-format pdf \
    --bootstrap 100
```

Figure	Shows
`dotplots/<taxon>.pdf`	Oxford-style synteny dot plot: linkage-group position against that taxon's own chromosomes. Fusion appears as two blocks on one row; mixing as interleaving.
`ideograms/<taxon>.pdf`	Each chromosome as a painted bar — the compact, publication-ready form of the same information.
`ribbon.pdf`	All taxa stacked, orthologs connected between adjacent tracks. Parallel ribbons = conserved synteny; crossings = rearrangement.
`character_matrix_heatmap.pdf`	The whole matrix as a coloured grid — the fastest way to spot missing data and near-constant characters.
`tree_*.pdf`	Cladogram per criterion with synapomorphies labelled on the branches where they originate.
`dotplots/<taxon>.wgd.pdf`, `ribbon_wgd/`	With `--highlight-wgd`: rescued ohnologs drawn at double diameter and ringed.

Order ribbon tracks by the tree

A ribbon plot connects only vertically adjacent tracks, so an arbitrary taxon order places unrelated genomes side by side and generates crossings that reflect nothing but that choice. `--ribbon-order-from-tree` uses a ladderized traversal so every clade is contiguous — removing ordering artefacts while leaving genuine rearrangement visible. Use `--ribbon-order` to show any subset of taxa, which is usually far more readable than all of them.

7.8 Example H — full sensitivity analysis

The recommended final step whenever thresholds were chosen rather than derived.

```
#!/usr/bin/env bash
set -euo pipefail
CORE="Amborella Vitis Oryza"

for minalg in 10 15 20 30; do
  python3 genomes_to_rbh.py \
      --input-dir ./Data --consensus-species $CORE \
      --consensus-min-species 2 --consensus-min-frac 0.66 \
```

```
      --consensus-min-alg-orthogroups "$minalg" \
      --threads 30 \
      --rbh-outdir "./rbh_t${minalg}" \
      --parsimony-script "$(pwd)/alg_parsimony.py" \
      --parsimony-outdir "./parsimony_t${minalg}" \
      --forward --min-genes 3 --alpha 0.05 \
        --outgroup Amborella --bootstrap 100
done

echo "=== topologies across thresholds ==="
for minalg in 10 15 20 30; do
  printf "%-8s " "t=$minalg"
  cat "./parsimony_t${minalg}/tree_reversible.nwk"
done
```

If the topology and the well-supported clades hold across the range, report your chosen threshold and move on. If they do not, that instability is a genuine result about how well-determined your character matrix is, and should be disclosed rather than resolved by picking the most attractive threshold.

8. Reading the output

8.1 From genomes_to_rbh.py

File	Contents
`<taxon>.rbh`	Ortholog table per taxon, with an `orthology_type` column distinguishing `direct` from `wgd_inferred` (rescued) hits.
`consensus_linkage_group_sizes.tsv`	Consensus mode: every candidate linkage group ranked by member-orthogroup count, with a `kept` column. The basis for choosing `--consensus-min-alg-orthogroups`.
`duplication_blocks.tsv`	With `--detect-wgd`: confirmed ohnologous chromosome pairs, supporting paralog-pair counts and p-values.

8.2 From alg_parsimony.py

File	Contents
`report.txt`	Everything below in one human-readable summary. Start here.
`character_matrix.tsv`	Taxa × characters, states 0/1/2/?.
`alg_scaffold_calls.tsv`	Every fusion/mixing call per taxon, with the evidence behind it.
`tree_reversible.nwk / .txt`	Best tree under unordered parsimony. With `--bootstrap`, support values are embedded as internal-node labels.
`tree_irreversible.nwk / .txt`	Best tree forbidding reversal (Camin-Sokal).
`synapomorphies_*.txt`	Which fusion/mixing events unite which clades.
`ci_ri_*.tsv`	Per-character consistency and retention indices. Always written.
`bootstrap_*.tsv`	Per-clade support fractions.
`permutation_test_*.tsv`	Per-replicate scores from the signal test.
`ohnolog_summary.tsv`	With `--highlight-wgd`: rescued vs. direct ortholog counts per taxon, chromosome and linkage group.

8.3 The three diagnostics

A most-parsimonious tree is a point estimate. These say how much to trust it.

- **CI / RI** (always computed). A consistency index below 1 flags a character that arose more than once independently — not the clean single-origin synapomorphy it appears to be in a synapomorphy list. Check these before quoting any individual character as evidence.
- **Bootstrap support.** Characters are resampled with replacement and the search rerun. Clades entailed purely by outgroup rooting are excluded, since no topology could fail to contain them. Macrosynteny matrices are small, so moderate support is common and worth reporting honestly.
- **Permutation test.** Asks whether the matrix beats randomly shuffled data of identical composition. A high p-value is an explicit warning that the topology should not be interpreted. Because these matrices typically have few characters, treat this as a precondition for interpreting any tree, not an optional extra.

Cross-check large analyses

Above `--max-exhaustive` taxa (default 8) the search is an NNI hill-climber with random restarts, not a validated general-purpose search — and bootstrap and permutation replicates inherit that limitation. Export with `--export-nexus` / `--export-tnt` and verify in PAUP* or TNT for anything consequential.

9. Troubleshooting

Symptom	Cause and fix
Hundreds or thousands of linkage groups	Usually `--consensus-min-frac 1.0` being too strict for a rearranged clade, or a fragmented core genome. Sweep min-frac; check contiguity (§3.3); then apply `--consensus-min-alg-orthogroups`.
Exactly 1 linkage group	min-frac far too permissive — unrelated groups are merging. Raise it. Also check `largest_group` in the sweep TSV.
Two different min-frac values give identical results	The threshold sits just above a k/n boundary and both round to the same requirement (§6.3). Check the k-of-n line Anvaya prints on every run.
"Input file seems to be empty" from diamond	`--min-scaffold-length` excluded every scaffold in a genome. Current versions exit with an explanatory message naming the species and its longest scaffold.
"--outgroup X not found among taxa"	X is the reference genome, which is deliberately not a taxon (§5.1). Pick a compared species, or switch to consensus mode where core genomes are taxa.
"--constraint mentions taxa not found"	Same cause, or the constraint was written for a larger dataset. `ls rbh_out/*.rbh` lists the valid taxon set.
"Could not find alg_parsimony.py"	Auto-detection failed — often because the calling script changed directory first. Pass `--parsimony-script "$(pwd)/alg_parsimony.py"` explicitly.
No fusion characters at all	Either the genomes genuinely retain the ancestral arrangement, or thresholds are too strict. Loosen `--min-genes` / `--alpha` and inspect `alg_scaffold_calls.tsv`.
A duplicated genome shows no linkage groups	The classic RBH dilution failure (§7.4). Enable `--detect-wgd --wgd-species …`.
duplication_blocks.tsv is empty	`--wgd-min-paralog-pairs` too high. Rerun with 1 to see the full distribution, then set it just above the noise floor.
0 ID-to-chromosome mappings extracted	Annotation-format mismatch — e.g. a GTF-style file named `.gff`. Inspect column 9: `ID=…` is GFF3, `gene_id "…"` is GTF. Rename to match.
Figures missing, no error	matplotlib not installed — plotting degrades gracefully with a warning rather than failing.

10. Complete flag reference

10.1 genomes_to_rbh.py

Input and output

Flag	Meaning
`--input-dir`	Directory of genomes: `{name}.cds` + `{name}.gff3|.gff|.gtf` (+ optional `{name}.pep`).
`--rbh-outdir`	Where to write `.rbh` tables (required).
`--species`	Restrict the run to these species plus the reference/core set [all discovered].
`--work-dir`	Scratch directory for DIAMOND databases and cached alignments.
`--rbh-only`	Stop after writing .rbh files; skip the parsimony step.
`--parsimony-script`	Path to alg_parsimony.py [auto-detected].
`--parsimony-outdir`	Output directory for alg_parsimony.py.
`--forward`	Everything after this is passed to alg_parsimony.py unchanged.

Linkage-group definition (choose exactly one mode)

Flag	Meaning
`--reference-species`	Single-reference mode: this genome's chromosomes are the linkage groups. Not itself a taxon.
`--consensus-species`	Consensus mode: reconstruct linkage groups from these ≥2 core genomes. All genomes become taxa.
`--consensus-min-species`	Minimum core genomes an orthogroup must appear in [all core species].
`--consensus-min-frac`	Fraction of core genomes that must agree on co-location before merging [1.0 = unanimous]. Only k/n values are achievable — see §6.3.
`--consensus-min-alg-orthogroups`	Discard reconstructed groups with fewer member orthogroups than this [1 = keep all].

Filtering and duplication

Flag	Meaning
`--min-scaffold-length`	Exclude genes on scaffolds shorter than this (bp) from the search entirely [0 = off].
`--detect-wgd`	Detect whole-genome/segmental duplication and rescue ohnologs that plain RBH drops.
`--wgd-species`	Restrict --detect-wgd to these species [all non-reference/non-core species].
`--wgd-min-paralog-pairs`	Minimum intragenomic paralog pairs to call two chromosomes ohnologous [5].
`--wgd-alpha`	Significance cutoff for the ohnologous-pair enrichment test [0.05].
`--wgd-output-dir`	Reuse translated .faa files from a prior run.

Search parameters

Flag	Meaning
`--evalue`	DIAMOND E-value cutoff [1e-3].
`--max-target-seqs`	DIAMOND hits retained per query [5].
`--threads`	DIAMOND threads [4].
`--genetic-code`	NCBI translation table for CDS→protein [1].
`--no-reuse-blast`	Recompute alignments even when a cached result with matching parameters exists [off; cached results are reused, which is what makes sweeps fast].

10.2 alg_parsimony.py

Input and character calling

Flag	Meaning
`--indir`	Directory of .rbh files, one per taxon (required).
`--outdir`	Output directory [alg_parsimony_out].
`--reference-prefix`	Override auto-detection of the shared ALG/reference column prefix.
`--alpha`	Significance cutoff for calling an ALG present on a chromosome [0.05].
`--min-genes`	Minimum shared orthologs to call an ALG present [3].
`--mixing-threshold`	Fraction of gene-order switches above which a fusion is “mixed” rather than “clean” [0.3].

Tree search

Flag	Meaning
`--outgroup`	Taxon rooting the irreversible analysis [auto: fewest derived states — verify this choice].
`--constraint`	Backbone constraint as a Newick string or file; any subset of taxa, polytomies allowed.
`--max-exhaustive`	Taxon-count ceiling for exact enumeration; above this a heuristic NNI search runs [8].
`--nni-restarts`	Random restarts for the heuristic search [10].
`--seed`	RNG seed for reproducible searches and replicates [1].

Support and diagnostics

Flag	Meaning
`--bootstrap N`	Bootstrap replicates; support embedded in the .nwk file and written to .tsv [0 = off].
`--bootstrap-restarts`	NNI restarts per bootstrap replicate (heuristic regime) [3].
`--permutation-test N`	Signal-test replicates: does the matrix beat shuffled data? [0 = off].
`--permutation-restarts`	NNI restarts per permutation replicate [3].

CI/RI homoplasy indices are always computed — no flag needed.

Figures

Flag	Meaning
`--plot-dotplots`	Oxford-style synteny dot plot per taxon.
`--plot-ideograms`	Chromosome-painting bars per taxon.
`--plot-ribbon`	Multi-track ribbon plot connecting orthologs between adjacent taxa.
`--ribbon-order`	Explicit taxon order for the ribbon plot; any subset of ≥2 taxa.
`--ribbon-order-from-tree`	Order ribbon tracks by a ladderized traversal of the `reversible` or `irreversible` tree.
`--plot-matrix`	Taxa × characters heatmap.

Flag	Meaning
`--plot-trees`	Cladogram per criterion with synapomorphies on branches.
`--plot-format`	pdf, png or svg [pdf].
`--highlight-wgd`	Extra dot plot, ribbon plot and summary table highlighting rescued ohnologs.

Interoperability

Flag	Meaning
`--export-nexus`	Write a NEXUS matrix with a PAUP* command block.
`--export-tnt`	Write a Hennig86 matrix with a TNT script (reversible criterion only).
`--run-paup / --run-tnt`	Also execute PAUP*/TNT if present on the system.
`--paup-bin / --tnt-bin`	Executable name or path [paup, tnt].

10.3 sweep_consensus_params.py

Flag	Meaning
`--input-dir, --consensus-species`	As for genomes_to_rbh.py (required).
`--outdir`	Directory for intermediate runs and the summary TSV [consensus_sweep].
`--out-tsv`	Path for the summary TSV [<outdir>/consensus_sweep_summary.tsv].
`--min-species`	Values of --consensus-min-species to sweep.
`--min-frac`	Values of --consensus-min-frac to sweep [1.0 0.75 0.5].
`--min-alg-orthogroups`	Values to report; derived arithmetically, so extra values are nearly free [1 5 10 15 20 30].
`--keep-runs`	Keep each combination's .rbh output directory.
`--extra`	Everything after this is passed to genomes_to_rbh.py unchanged.

A closing note on honest reporting

Several of Anvaya's thresholds have no objectively correct value, and the software is built to make that visible rather than hide it — the sweep TSV, the size distributions, the k-of-n warnings and the three diagnostics all exist so choices can be defended or disclosed. Where a threshold was chosen rather than derived, say so, and report whether the conclusions survive reasonable alternatives. That is more useful to a reader than a single tidy number.